SVM Primal Problem & Quadratic Equation in α with Constraints

Minimize: $\frac{1}{2}\,\alpha^{n}P\alpha + q^{n}\alpha$ **Subject to:** $\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4 = 0$ $0 \le \alpha_1, \alpha_2, \alpha_3, \alpha_4 \le 1$ $\alpha_1^2 + 2\alpha_1\alpha_2 - 5.5\alpha_1\alpha_3 - 6\alpha_1\alpha_4 + 2\alpha_2\alpha_1 + 4\alpha_2^2 - 11\alpha_2\alpha_3 - 12\alpha_2\alpha_4 - 5.5\alpha_3\alpha_1 - 11\alpha_3\alpha_2 + 32.5\alpha_3^2 + 36\alpha_3\alpha_4 - 6\alpha_4\alpha_1 - 12\alpha_4\alpha_2 + 36\alpha_4\alpha_3 + 40\alpha_4^2$

Quadratic Equation in α with constraints

Minimize: $\frac{1}{2}$ [$2\alpha_1^2 + 4\alpha_1\alpha_2 - 11\alpha_1\alpha_3 - 12\alpha_1\alpha_4$

$+ 4\alpha_2\alpha_1 + 8\alpha_2^2 - 22\alpha_2\alpha_3 - 24\alpha_2\alpha_4$

$- 11\alpha_3\alpha_1 - 22\alpha_3\alpha_2 + 65\alpha_3^2 + 72\alpha_3\alpha_4$

$- 12\alpha_4\alpha_1 - 24\alpha_4\alpha\alpha_2 + 72\alpha_4\alpha_3 + 80\alpha_4^2$] - [$\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4$]

$$P = \begin{bmatrix} 2 & 4 & -11 & -12 \\ 4 & 8 & -22 & -24 \\ -11 & -22 & 65 & 72 \\ -12 & -24 & 72 & 80 \end{bmatrix}$$

$q^{n} = [\,1 \;\; 1 \;\; 1 \;\; 1\,]$